- Provide few web pages developed using HTML only and ask student groups to improve them using CSS

Quality inputs:

- Computers with simple text editors, web browser, presentations
- Requirement descriptions

Competency Level 10.6: Uses an authoring tool to create web pages

Time: 10 periods

Learning Outcomes:

- Briefly explains web authoring tools
- Creates web pages using a web authoring tool

Contents:

- Introduction to web authoring tools

Concepts and terms to be highlighted:

- Appreciate that web pages can be generated using web authoring tools (standard development software) without writing html tags
- Appreciate that web pages can be further be developed using html tags
- Web authoring tools
    - Ex: - Dream weaver, Composer, etc.

Guidance for lesson plans:

- Divide the students into groups
- Demonstrate how to generate web pages using a suitable web authoring tool
- Demonstrate that generated web pages can be edited by manually with HTML viewer/editor of that tool
- Ask the student groups to develop web site using authoring tool
- Ask student groups to edit the generated web site and make some changes

Guidance for assessments and evaluations:

- Ask the student groups to generate some web pages using a web authoring tool specified by the teacher (If Internet access is available, the teacher may guide the students to use an online tool)
- Ask the student groups to further develop them using html tags
- Ask the student groups to further develop them using CSS
- Ask the student groups to link the pages appropriately and make a comprehensive web site

Quality inputs:

- Computers with Web authoring tool, web browser
    - presentations
    - Requirement descriptions
    - Internet facilities

Competency Level 10.7: Creates dynamic web pages using PHP and MySQL

Time: 06 periods

Learning Outcomes:

- Defines dynamic web pages
- Creates data source and enters data
- Creates PHP code to save/retrieve data to and from MySQL
- Develop simple web based information systems

Contents:

- Introduction to dynamic web pages
- Embedding PHP code into web page
  - Variables
  - Arrays
  - Control structures
  - Functions
  - Database connectivity
  - Working with databases
- Forms
  - Input element
    - ❖ Type attribute
    - ❖ Name attribute
    - ❖ Value attribute
  - Text input (Password)
  - Radio buttons
  - Check box
  - Selection
  - Submit buttons
  - Reset button
  - Action attribute
  - Method attribute
    - ❖ Get
    - ❖ Post
  - Grouping form data using <field set> tag
  - Saving form data into database
- Creating data source and entering data
- Creating PHP code to retrieve data from MySQL database
  Set form values using retrieved data

Concepts and terms to be highlighted:

- Characteristics of a dynamic web page
- Difference between the server based scripting and desktop applications
- Need for the use of a server side scripting language to access a database
- Embedding PHP code into a web page
- Concept of web based information systems
- Appreciate how data source (MySQL) is accessed using a server side scripting language (PHP) via a web page

- Web browser
- Scripting language (Example: PHP)
- Server (Example: Apache)
- Database software (Example: MySQL)
- WAMP, XWAMP

Guidance for lesson plans:

- Divide the students into groups
- Demonstrate few static web pages and dynamic web pages
- Ask students to identify the features of static web pages and dynamic web pages (Example: data source available in dynamic etc.)
- Introduce PHP (commands, syntax etc.)
- Demonstrate how to create a data source (MySQL) and populate the database with some data using MySQL commands
- Demonstrate how to embed an appropriate PHP code in to the source web page to retrieve data from the MySQL database
- Demonstrate how to embed an appropriate PHP code in the source web page to enter and save data in the MySQL database and view the data
- Ask student groups to develop a simple web based information system

Guidance for assessments and evaluations:

- Create a data source (MySQL) and populate the database with some data using MySQL commands
- Create a web page and embed an appropriate PHP code into it to retrieve data from the MySQL database
- Embed an appropriate PHP code in the source web page to enter and save data in the SQL database and view the data
- Demonstrate the system to the class

Quality inputs:

- Computers with Server software (ex:- XAMP/WAMP/LAMP)
- Internet and Internet resources (ex:- YouTube)
- A list of required PHP and MySQL commands
- Presentations

Competency Level 10.8: Publishes and maintains web sites

Time: 04 periods

Learning Outcomes:

- Publishes the developed website locally
- Identifies free web hosting sites from the Internet
- Publishes the developed website through a free web hosting site
- Investigates the factors affecting performance of website

Contents:

- Local publishing
  - Own computer, intranet
- Internet publishing
  - Connecting to the web Service provider
  - Publishing web Pages on web server
- Factors affecting performance of website

Concepts and terms to be highlighted:

- Appreciate the need of publishing of developed website, locally and globally
- Identify the requirements of local publishing
  - Conform to the specification
  - Conform to the quality - testing the interface and appearance, authentications, web browser compatibility, etc.
- Identify the procedures of Internet publishing
  - Ex: The need of ISPs and Protocols etc.
- Appreciate different ways of web hosting
  - Ex : free & paid, and shared & dedicated web hosting
- Identify few web hosting sites
- Appreciate the need for maintaining and upgrading the website for uninterrupted browsing and keeping the information up to date
- Identify the factors affecting (upload/download speeds etc. as well as usability, navigation, concurrent use etc.) the performance of the web site
- Appreciate the need of search engines making DNS servers updated (SEO - Search Engine Optimization) with the new websites
- Appreciate the need for protecting the website against hackers

Guidance for lesson plans:

- Visiting some websites and discuss the concept of publishing (locally and globally, separately) so that other people can visit the site and read the content
- Divide the students into groups
- Ask student groups to host the website they developed in the local server (or desktop) and check against the specification whether the requirements and quality is achieved, and present their experience
- Discuss that the local publishing is necessary as a test before global publishing

- Ask the student groups to find the procedures (creating a unique name, uploading the website etc.) of global publishing through and Internet search, and discuss.
- Ask student groups to find some web hosting sites (both free and paid) through an Internet search
- Ask student groups to host their web site in a free hosting site following the procedures they discovered
- Discuss the need for maintaining and upgrading the website for uninterrupted browsing and keeping the information up to date
- Discuss and ask the student groups to identify the factors affecting (upload/download speeds etc. as well as usability, navigation, concurrent use etc.) the performance of the web site
- Discuss the need and the procedures of updating DNS using web crawler
- Discuss the need and the procedures of Search Engine Optimization (SEO)
- Discuss the need for protecting the website against hackers and ask the students to find some interesting cases using an Internet search
- Discuss how access is controlled using passwords and the content is protected using encryption techniques (Briefly mention about http and https and demonstrate few websites using https – bank transactions, credit card transactions etc.)

Guidance for assessments and evaluations:

- Divide the students into groups
- Ask the groups to host their developed web site in a free hosting site following the procedures they studied

Quality inputs:

- Internet facility, Presentations, LAN facility, Software for file uploading(ex: filezilla)

Competency 11: Explores IoT and identify the building blocks of embedded systems to develop simple applications

Competency Level 11.1: Acquires the knowledge of basic building blocks of embedded systems

Time: 08 periods

Learning Outcomes:

- Identifies and lists microcontroller based development boards
- Describes available features on a microcontroller based development board
- Identifies necessary software and download them from the Internet to develop a microcontroller based embedded system
- Develops simple applications using a microcontroller based development board
  - Switch a LED on/off on ambient light intensity
  - Run a fan on room temperature
  - Detect door opening /closing using a read switch

Contents:

- Microcontroller - Based Development Systems (Arduino and other similar systems)
- Introduction
  - Embedded systems
  - Microcontroller based embedded systems vs. microprocessor based embedded systems
- Features
  - Analog Input
  - Digital Input
  - Microcontroller
  - Digital Output
  - Data Receiver and Transmitter
  - Communication Port
  - Power supply
- Connect to the computer
  - USB Connectivity
  - IDE Software (code editor, compiler and programmer)
- Simple applications
  - Switching a LED on/off
  - Sensing ambient light intensity with a LDR and switching a LED on and off depending on the light intensity
  - Sensing the room temperature with a temperature sensor and switching a fan on and off depending on the temperature
  - Detection of opening/closing a door using a read switch

Concepts and terms to be highlighted:

- Embedded system - typically a microcontroller based system embedded into an external electro-mechanical system (known as Physical Computing)
- Microcontroller based embedded systems vs. microprocessor based embedded systems
- Microcontroller - a single chip with CPU, memory, input/output ports, and other hardware components (Timers, Counters, Oscillators, Analog - Digital Converter etc.)
- Embedded system
- Microcontroller
- Physical Computing
- Features (Analog Input, Digital Input, Digital Output, Data Receiver and Transmitter, Communication Port, Power supply, Reset switch)

Guidance for lesson plans:

- Discuss the similarities and differences between a general purpose computer and an embedded system (compare a general purpose computer with digital camera, mobile phone, a computerized car etc.)
- Divide the students into groups
- Provide a microcontroller based development board to each student group and ask to identify features in the development board
- Describe the purpose and the functionality of each feature
- Demonstrate a sample embedded system and its functionality
- Discuss and identify a suitable physical system for which an embedded system can be developed
- Ask the student groups to identify the required components for the proposed embedded system
- Guide the student groups to draw a schematic diagram for the proposed embedded system
- Guide the student groups to assemble the components of the proposed embedded system according to the schematic diagram
- Discuss the need to develop a software for controlling the embedded system assembled using an Integrated Development Environment (IDE)
- Demonstrate how to use a Code Editor, Compiler and Programmer integrated into an IDE
- Guide the student groups to design the software to be programmed into the microcontroller starting from a flowchart
- Introduce the syntax of the selected programming language for the purpose
- Guide the student groups to transform the flowchart into a source code using the IDE
- Guide the student groups to compile the source code and program the machine code into the microcontroller using the IDE
- Guide the student groups to connect the embedded system to a power supply and observe the intended behavior of the system (observe the outputs for specific inputs and verify that the required functionality is delivered)
- Ask the student groups to design, code and implement the following embedded systems:
    - Switch a LED on and off

- Sensing ambient light intensity with a LDR and switching a LED on and off depending on the light intensity
- Sensing the room temperature with temperature sensor and switching a fan on and off depending on the temperature
- Detection of opening/closing a door using a read switch

Guidance for assessments and evaluations:

- Provide a scenario to the student groups to develop an embedded system
- Ask student groups to develop the embedded system according to the steps that they have learnt
- Organize an exhibition of embedded systems developed in the class
- Ask student groups to demonstrate the embedded systems developed

Quality inputs:

- Open source electronic platform - Microcontroller based development boards (Arduino, Micro:bit, Raspberry pi)
- Input/output components (Example: LDR for input, LED for output)
- Computers with python software (Integrated Development Environment - IDE)
- Internet connection
- Presentations